

CHAPTER

0

Contents

I Precalculus

1	Functions	1
1.1	Function basics	1
1.1.1	Understanding functions	1
1.1.2	More examples of functions	3
1.1.3	Exercise	3
1.2	Function composition and inverses	7

CONTENTS

1.2.1	Exercise	10
1.3	Injection, surjection and bijection	12
1.3.1	Exercise	15
1.4	An aside: A more rigorous definition of a function	17
2	Linear Algebra	19
2.1	Vector Basics	19
2.1.1	Exercise	24
2.2	Scaling and dot product	26
2.3	Matrices	31
2.4	Determinants, inverses and Cramer's rule	38
2.4.1	Exercise	42
2.5	Cross products and triple products	45

Part I

Precalculus

CHAPTER

1

FUNCTIONS

1.1 FUNCTION BASICS

1.1.1 Understanding functions

Before we start on the calculus topics, this part of the book will acquaint you with the mathematics you would need to understand in order to read this book. If you are familiar with these topics, you may skip to the Part II on Calculus I. For the rest of you, the pace of this part will be very fast, it is recommended that you read up on the concepts

you do not understand very well. We will start with the idea of a *function*. By now, I am sure that many of you may have seen equations like these:

$$x^2 - 5x + 6 = 0 \quad (1.1.1)$$

$$\sin(x) = 0.5, 0 \leq x \leq \frac{\pi}{2} \quad (1.1.2)$$

$$y = mx + c \quad (1.1.3)$$

Out of the three equations, notice that Equation (1.1.3) is special. There is not some fixed x values that you must plug in. Instead, the y value varies with the x value. Given a x value, the equation “produces” a y value. This brings us to the idea of a function. A function is simply something that takes in a value, and “produces” another value based on the input. You may have come across functions before, such as the trigonometric functions like $\sin$ and $\cos$. The value of $\sin(x)$ varies with different values of x . Of course, it is time for me to give you the definition of a function.

Definition 1.1 Function

A **function** f is a mathematical object that assigns its inputs, or *domain*, to its outputs, or *range*. The notation $f : A \rightarrow B$ means that the function has a domain A and range B . The notation $f(a)$ where $a \in A$ denotes the element of B to which a is assigned to.



Note that A and B are sets.

Example 1.1 Evaluating functions

Here are some examples of functions:

$$f(x) = x + 1 \quad (1.1.4)$$

$$g(x) = \sin(x) \quad (1.1.5)$$

$$h(x) = x^2 - 5x + 6 \quad (1.1.6)$$

To evaluate the output of a function, you simply substitute the input to the expression that defines the function. Notice that for Equation (1.1.4), it is simply the line $y = x + 1$. Each value of $f(x)$ for some $x \in \mathbb{R}$ is just y . Take $x = 1$ for example, $f(1) = 1 + 1 = 2$ and for the line $y = x + 1$, notice the y value here is also 2. The function assigns each x value to $x + 1$. Hence, in this case, $f : \mathbb{R} \rightarrow \mathbb{R}$. For Equation (1.1.5), the function takes any real x value and maps it to $\sin(x)$. As a result, its domain would be $\mathbb{R}$ and its range would be $[-1, 1]$. Again, it is just

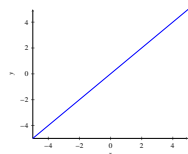


Figure 1.1: Graph of $y = x$.

like the graph $y = \sin(x)$. Lastly, the quadratic in Equation (1.1.6) is another real function (i.e. it has a real domain and range). Think about what happens when $h(x) = 0$, what value of x must you plug in to make $h(x) = 0$. Well, isn't it just the roots of the quadratic. Hence, the roots of a function are just the values x such that $h(x) = 0$. In this case, it would be 2 and 3. Thus, $h(2) = h(3) = 0$.



1.1.2 More examples of functions

Actually, functions can be more than the usual $f(x) = \text{something}$. Let's say you want to have a function that outputs "1" if the input is rational and "0" if it is not, you can define it as such:

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases} \quad (1.1.7)$$

This is actually the famous Dirichlet function that shatters certain intuitions on functions and will be discussed in later chapters.

Here, $f(\frac{1}{2})$ would be 1 but $f(\sqrt{2})$ would be 0. (You would probably get a migraine from graphing it.) This is an example of having conditionals in a function. Lastly, functions may take more than one input and produce more than one output, examples include:

$$f(x, y) = x^2 + y^2 \quad (1.1.8)$$

$$g(t) = (\cos(t), \sin(t)) \quad (1.1.9)$$

Equation (1.1.8) is an example of a *multivariate* function, in other words, it takes in multiple inputs. If $x = 2$ and $y = 2$, then $f(2, 2) = 8$. On the other hand, Equation (1.1.9) does not have a single output. Instead, it has a tuple as its output. Evaluating it at $t = \pi$, notice that the output is $(-1, 0)$. What is interesting is that if you trace out its outputs on a plane for all $t \in \mathbb{R}$, you will get:

As you will see in Part IV of the book, this is the parametric equation of a circle. A tuple is basically a set where order matters. You can treat it as if it is a coordinate.

Functions are the building blocks of mathematics, and this book will deal extensively with it. Therefore, it is paramount that you get the basics right. Try the following exercise:

1.1.3 Exercise

Problems

1. Given that $x = 2$ and $y = \pi$, evaluate $f(x)$ for:

(a) $f(x) = x^3 + 1$

(b) $f(x) = 2^x$

(c) $f(x) = \begin{cases} x^4 + 1 & \text{if } x < 2 \\ 3^x & \text{otherwise} \end{cases}$

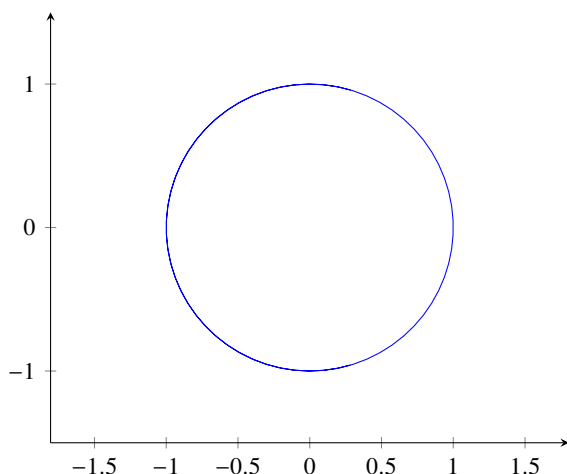


Figure 1.2 Plot of Equation (1.1.9) for all $t \in \mathbb{R}$

(d) $f(x, y) = \frac{\sin(y)}{x}$

2. Given the following functions, write down $f : A \rightarrow B$ where A is the domain and B is the range (do not consider complex inputs and outputs, we shall be ignorant to it for the time being to simplify things):

(a) $f(x) = x$

(b) $f(x) = \frac{1}{x}$

(c) $f(x) = \sqrt{x}$

3. Find the root(s) of the following functions:

(a) $f(x) = x^2$

(b) $f(x) = 2^x - 1$

(c) $f(x) = \sin(x)$

4. A *recursive* function is one where the function uses itself in its definition. For example, the Fibonacci sequence can be defined like this for any Fibonacci number n :

$$f(n) = \begin{cases} f(n-1) + f(n-2) & n > 1 \\ 1 & n = 1 \\ 0 & n = 0 \end{cases}$$

*This function has a special name, the **identity** function. Notice how its input and output are the same.*

- (a) Compute $f(3)$ explicitly using the definition of f .
 - (b) Define the factorial of a positive integer in a similar way using recursion, define it as $g(a)$.
 - (c) Sketch the graph of $g(a)$.
5. Let $f(x)$ be a polynomial. If $f(a) = 0$, prove that $x - a$ is a factor of f .

Solutions

1. (a) 9
(b) 4
(c) 9
(d) 0
2. (a) $f : \mathbb{R} \rightarrow \mathbb{R}$
(b) $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ (Note the function is not defined at $x = 0$)
(c) $f : \mathbb{R}_0^+ \rightarrow \mathbb{R}_0^+$ (Square root not defined for negative numbers)
3. (a) 0
(b) 1
(c) $n\pi, \forall n \in \mathbb{Z}$
4. (a) $f(3) = f(2) + f(1) = f(1) + f(0) + 1 = 2$
(b) $g(a) = \begin{cases} ag(a-1) & a > 0 \\ 1 & a = 0 \end{cases}$
(c) Because our definition of $g(a)$ is discontinuous (the function have “gaps”), we shall only plot the points where the function is defined.

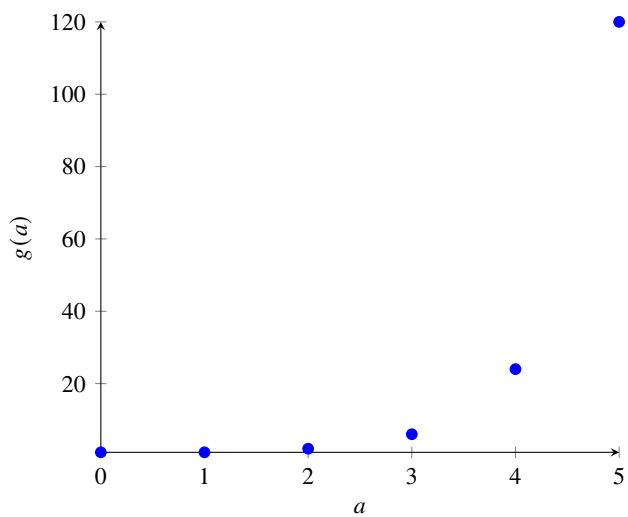


Figure 1.3 Graph of $g(a)$.

1.2 FUNCTION COMPOSITION AND INVERSES

In this chapter, we will look at composition of functions and inverse functions. You can use the normal arithmetic operations on functions like how you would to numbers:

Example 1.2 Arithmetic operations on functions

If $x = 2$ and $f(x) = x^2 + 4$:

$$f(x) + f(x) = 2^2 + 4 + 2^2 + 4 = 16$$

$$\frac{f(x)}{2} = \frac{2^2 + 4}{2} = 4$$



What is more interesting, perhaps, is taking the *output* of a function and use it as the *input* to another function. You may also think of it as “piping” one function into another. This is called *function composition*.

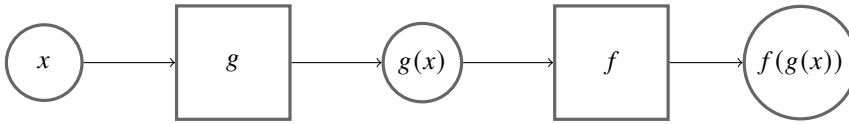


Figure 1.4 Visual representation of function composition.

Another notation for composing functions together (which we will not really use except for this chapter) is as such:

$$f(g(x)) = (f \circ g)(x)$$

Note that the rectangles are functions and circles are values.

When you compose the same function n -times, the notation for it is:

$$f^n(x) = \underbrace{f(f(\dots f(x)))}_{n\text{-times}}$$

However, this does not apply to trigonometric functions. $\sin^2(x)$ is $\sin(x) \times \sin(x)$ instead of $\sin(\sin(x))$. Annoying mathematical notation strikes again. What is even better, is that $f^{-1}(x)$ denotes the inverse of f .

Example 1.3 Composing functions

Here are some examples of composing functions together: If $x = 3$ and $f(x) = \pi x$, $g(x) = \sin(x) + 1$:

$$g(f(x)) = \sin(\pi \times 3) + 1 = 1 \quad (1.2.1)$$

$$f^3(x) = \pi^3 x \quad (1.2.2)$$

There is nothing much here, really, just evaluate the function from the innermost function to the outermost.

**Theorem 1.1** Composition of functions is associative

Given functions $f : X \rightarrow Y$, $g : Y \rightarrow Z$ and $h : Z \rightarrow W$, prove that the composition of the three functions is *associative*, in other words:

$$h \circ (g \circ f(x)) = (h \circ g) \circ f(x)$$

QED**Proof 1.1** Composition of functions is associative

Let $x \in X$, we have:

$$h \circ (g \circ f(x)) = h(g \circ f(x)) = h(g(f(x))) = h \circ g(f(x)) = (h \circ g)f(x)$$

This proves the associativity of function composition.

QED**Remark 1.1**

This is just a fun little exercise for the reader and ultimately is not that important to the entire book, just a good identity to keep in the back of your mind.



Next, we will delve into inverse functions. Let us start with one I believe you may have seen before, look at the following figure:

If I ask you what is $\arcsin(B)$, you may say it is $\frac{B}{C}$. But, what is $\arcsin(\sin(B))$, well, that is just B , isn't it. Hence, the arcsin sort of "cancels" the sin. We call arcsin the *inverse function* or just inverse of sin. Generally, given a function $f : X \rightarrow Y$, f^{-1} denotes the

We will use arcsin as the inverse of sin instead of $\sin^{-1}$ as it is clearer.

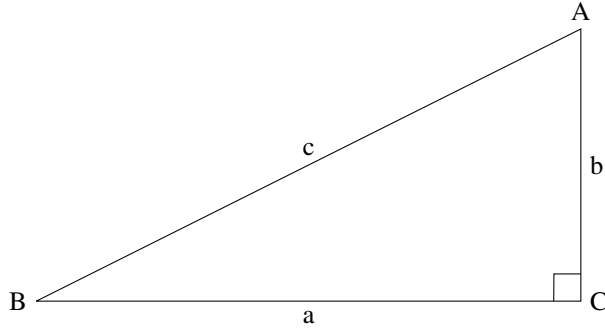


Figure 1.5 Figure of right triangle ABC.

inverse function that satisfies:

$$f \circ f^{-1}(y) = y, y \in Y \text{ and } f^{-1} \circ f(x) = x, x \in X \quad (1.2.3)$$

Note 1.1 Cancelling functions and their “inverse”

One common misconception is that $f(f^{-1}(x)) = f^{-1}(f(x))$. While this is true, you may apply this fact wrongly. For example, take the following function:

$$f(x) = \sin(x), f : \mathbb{R} \rightarrow \{y : -1 \leq y \leq 1\}$$

This looks deceptively simple, when asked to evaluate $f^{-1}(f(\pi))$, it is π (You may want to verify it yourself). But what is $f(f^{-1}(\pi))$? Well, according to our definition of f , $f^{-1}(\pi)$ is simply not defined. In actuality, to find $f^{-1}(\pi)$, we need complex numbers, which is not allowed in the definition. Hence, you must be extra careful when you are dealing with function inverses and pay attention to the domain and range.



To find the inverse function of a function f , we have to find the function that maps the outputs of the f back to its inputs. To solve algebraically, we set $f(x)$ for some arbitrary x to y , and make x the subject of the equation.

Example 1.4 Finding the inverse of a function

Given that $f(x) = \frac{x+7}{3}$:

$$\begin{aligned} f(x) &= y \\ \frac{x+7}{3} &= y \\ x+7 &= 3y \\ x &= 3y-7 \end{aligned}$$

Hence, the inverse function is $f^{-1}(y) = 3y - 7$.

**Example 1.5 Common inverse functions**

Some common examples of inverse functions:

1. Trigonometry:

arcsin arccos arctan ...

2. Logarithms (Inverses of exponential functions):

ln $\log_b$

**1.2.1 Exercise****Problems**

- Given $f(x) = \frac{x^2+3}{2}$ and $g(x) = \tan(x) + 1$, evaluate:
 - $f(g(\frac{\pi}{4}))$
 - $f^3(1)$
 - $f(g(\arcsin(0)))$
- Given that $h(x) = (x-7)^2$, $x \geq 7$, find the inverse of h .
- Hence, evaluate:
 - $h^{-1}(1)$

(b) $h(h^{-1}(69))$ As Peter Lorre would have said it, “Do it ze kveek vay, Johnny”.

Solutions

1. (a) $\frac{7}{2}$

(b) $\frac{61}{8}$

(c) 2

2. $h^{-1}(y) = \sqrt{y} + 7$

3. (a) 8

(b) 69 (Inverse function and the original function “cancels”.)

1.3 INJECTION, SURJECTION AND BIJECTION

You may have noticed in the previous exercise that question 2 is particularly interesting. There is a restriction on the domain of $h(x)$. Why is that? This is what this chapter is all about. However, before we explain that, I will go over some definitions of properties of functions. Firstly, we have *injection*.

Definition 1.2 Injective functions

An injective function, f , or a *one-to-one* function, is a function such that every distinct output of the function corresponds to only *one* distinct input, that is:

$$f(x_1) = f(x_2) \text{ implies } x_1 = x_2$$



By contraposition,
 $x_1 \neq x_2$ implies
 $f(x_1) \neq f(x_2)$. The
 contraposition of a
 statement “ p then q ”
 is “not q then p ”.

Of course, this explanation might not have been very clear. Here is a visualisation:

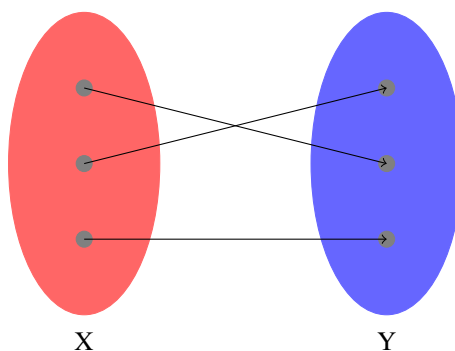


Figure 1.6 Illustration of an injective function with domain X and range Y .

As you can see in Figure 1.6, every output in Y of the function is associated with only one input in X , hence the function is *injective*. (Kind of like how marriages *should* be.) To prove that a function f is injective, you can either show that if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$, or if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

Here, the function is *not* injective. As you can see in Figure 1.7, the output y is associated with more than one input in X . With this in mind, it is time to talk about *surjection*.

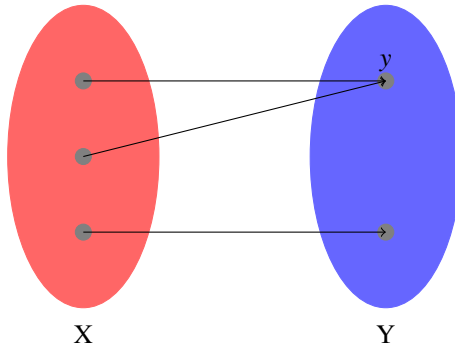


Figure 1.7 Illustration of a non-injective function with domain X and range Y .

Definition 1.3 Surjective function

A surjective function f is a function such that for every output y , there exists at least one input x such that $f(x) = y$.



From Figure 1.6 and Figure 1.7, both functions are surjective as every output is associated with at least one input. Do take note that it does not matter how many inputs associate to one output, as long as every output is associated with at least one input, the function is surjective. To prove that a function is surjective, you must show that there exists an element in the domain of the function for any arbitrary output in the range.

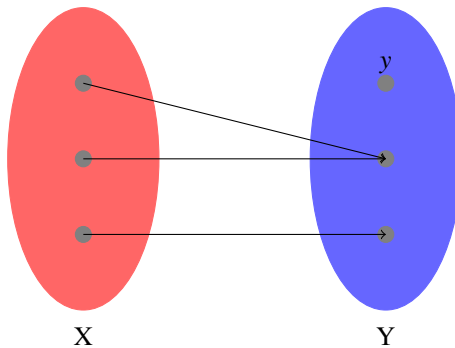


Figure 1.8 Illustration of non-surjective function with domain X and range Y .

From Figure 1.8, we can see that y has no input associated with it. As a result, the

function here is not surjective. Lastly, *bijective* functions can't be any easier.

Definition 1.4 Bijective function

A bijective function is a function that is both injective and surjective.



However, bijective functions have a superpower. A function is only bijective *if and only if* it has an inverse.

P iff Q, or "P if and only if Q", means that "if P then Q" and "if Q then P".

Theorem 1.2 Bijection and inverses

A function, $f : A \rightarrow B$, has an inverse if and only if it is bijective.

QED

Proof 1.2 Bijection and inverses

To begin such proofs where there is a "if and only if" clause, we will prove both cases separately. First, we claim that if the function has an inverse, then it is bijective. To prove this, we simply prove that f is injective and surjective (See Section 1.3). Then, we claim that if a function is bijective, then it has an inverse (See Section 1.3). With both proofs, we show that f has an inverse if and only if it is a bijection.

QED

Lemma 1.1 Inverse function has bijections

If a function, $f : A \rightarrow B$, has an inverse, then it is bijective.

QED

Proof 1.3 Inverse functions has bijections

Suppose f^{-1} is the inverse of f . Notice that $f^{-1}(f(x))$ is injective as it is simply $f(x) = x$. Suppose $a, a' \in A$ and $f(a) = f(a')$, then $f^{-1}(f(a)) = f^{-1}(f(a'))$. Notice that by injectivity of $f^{-1}(f(x))$, $a = a'$. Hence, f is injective. Then, suppose $b \in B$, we wish to show that the function is surjective. Also, $f(f^{-1}(x))$ is surjective(it is the identity function). Let $f^{-1}(b) = x$ and $f(f^{-1}(b)) = c$ for some $c \in B$. By a simple substitution, $f(f^{-1}(b)) = f(x) = c$, thereby showing

there exists a $x \in A$ for every $c \in B$. The two cases proves that f is bijective if it has an inverse.

QED

Lemma 1.2 Bijective function has inverses

If a function, $f : A \rightarrow B$, is a bijection, then it has an inverse.

QED

Proof 1.4 Bijective function has inverses

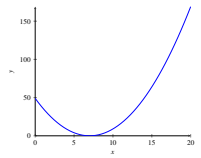
Let us define f^{-1} as $f^{-1} : B \rightarrow A$. Let $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Let $f^{-1}(b) = a$. Since f is injective, f^{-1} is well-defined, or a function that outputs the same value given the same input. Next, we will show that f^{-1} is indeed the inverse of f by showing the composition of the two is an identity function. Again, let $a \in A$ and $b = f(a)$ which implies, by definition, $f^{-1}(b) = a$. Composing the two functions, $f(f^{-1}(b)) = f(a) = b$, which shows that f^{-1} is indeed the inverse function. The other case of $f^{-1} \circ f$ is similar. Hence, if f is a bijection, then it has an inverse.

QED

Remark 1.2

This serves as a very good example showing how the reader should prove the injectivity, surjectivity, and bijectivity of functions. Also, this explains why in the previous exercise, the function $h(x) = (x - 7)^2$ is bounded, only taking x values larger than 7. The function is not bijective in $\mathbb{R}$, as evident in the graph as it is not injective. Hence, the bounds are needed for h to have an inverse.

△



1.3.1 Exercise

Figure 1.9: Plot of $(x - 7)^2$.

Problems

1. Determine if the following functions are injective or surjective or bijective in its domain:

(a) $f(x) = \sqrt{x}, f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$

(b) $f(x) = \frac{1}{x}, x > 0$

(c) $f(x) = (x - 7)^2, x \geq 5$

2. *Prove that:*

(a) The composition of two injective functions is *injective*.

(b) The inverse of a function is *injective*.

(c) The inverse of a function is *surjective*.

Solutions

1. (a) Bijective

(b) Injective

(c) Surjective

2. (a) Let the two functions be $f : A \rightarrow B$ and $g : B \rightarrow C$. If $g(f(x_1)) = g(f(x_2))$, by injectivity of g , $f(x_1) = f(x_2)$ and by injectivity of f , $x_1 = x_2$. Thus, $g \circ f$ is injective.

(b) Let $f : A \rightarrow B$ and f^{-1} be its inverse. Notice if $f^{-1}(x_1) = f^{-1}(x_2)$, then $f(f^{-1}(x_1)) = f(f^{-1}(x_2))$ which implies $x_1 = x_2$. Hence, f^{-1} is injective.

(c) Similarly, define f and f^{-1} . Let $y \in A$ and $f^{-1}(x) = y$. Notice that $x = f(y)$ and because f is surjective (by Section 1.3), $x \in B$, thus proving that f^{-1} is surjective.

And this concludes the chapter on functions. Readers are encouraged to do more similar practices to strengthen their knowledge on functions. Also, this chapter is more on the “pure mathematics” side, such proofs are only presented as a good exercise to the reader. In actuality, the book does not focus that much on this kind of mathematics (i.e. analysis) but more on the practical aspects of Calculus.

1.4 AN ASIDE: A MORE RIGOROUS DEFINITION OF A FUNCTION

TBD

CHAPTER

2

LINEAR ALGEBRA

2.1 VECTOR BASICS

Firstly, good job on finishing chapter 1, it is not an easy feat. Stepping into this chapter, you may feel overwhelmed by the name, *Linear Algebra*. What is that? Please take note though, the knowledge in this chapter will not be used until Calculus III to IV in part IV of the book. If you are just willing to learn Calc I and II, you do not need the knowledge here. With that said, let's delve into Vectors. A concept that is very ubiquitous in this world is a coordinate, like the GPS coordinate in the maps app. A

coordinate simply describes the position of a point in space. Generally, it is denoted (x, y) in 2 dimensions and (x, y, z) in 3 dimensions. Likewise, vectors also behave like coordinates and are written down in a *similar* fashion. Think of a vector as a coordinate.

Likewise, a vector at (x, y, z) is denoted $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Definition 2.1 Vector

A vector is a mathematical object with a magnitude, or length, and a direction. A vector, $\vec{a}$ is a point (x, y) in space drawn with an arrow. It is denoted as:

$$\vec{a} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Of course, vectors by themselves are not very useful. Hence, we shall define the vector space. For the purposes of this text, you may think of vector spaces as the 2 and 3 dimensional euclidean spaces you are very familiar with. In a vector space, we can finally meaningfully manipulate them. Take a look at the vector $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and the vector $\vec{b} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ in the two dimensional euclidean space:

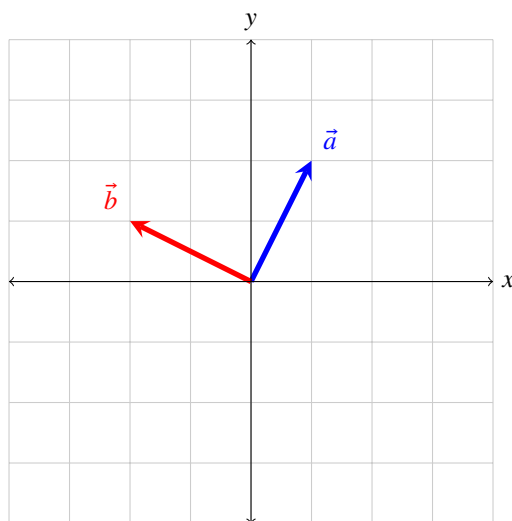


Figure 2.1 Figure of vectors $\vec{a}$ and $\vec{b}$.

You may have noticed that the coordinate axes in Figure 2.1 are the same as the ones you are familiar with. However, in linear algebra, instead of the usual xy axes to describe

the components of a vector, you have $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ to describe the components of a vector. (So instead of saying the “x” component of a vector, you would say the “i hat” component of a vector.) This is a convention in linear algebra. $\hat{\mathbf{i}}$ translates to the x axis and $\hat{\mathbf{j}}$ translates to the y axis. Likewise, in the third dimension, $\hat{\mathbf{k}}$ represents the z axis. Another thing to note are the *basis vectors*:

Some authors use $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ instead of the usual $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$. We shall use the former.

Definition 2.2 Basis vectors

The set of vectors in a vector space are called *basis* if every vector in the vector space can be written uniquely written as a finite *linear combination*¹ of the basis vectors. An example would be these vectors $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in the 2 dimensional euclidean space which can be thought of one unit of distance in the x and y direction respectively.

¹ A linear combination of x and y is $ax + by$ where a, b are constants.



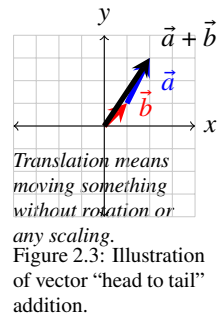
With this in mind, another way of writing a vector $\vec{a} = \begin{bmatrix} x \\ y \end{bmatrix}$ is $\vec{a} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$. Notice that like in Section 2.1, $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are written in a linear combination of x and y . Of course, these are not the only basis vectors in 2d, if you have learned polar coordinates before, there is the equivalent $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$, but this is a topic for a later chapter. Next, we shall look at the *addition* of vectors. A vector $\vec{a} = \begin{bmatrix} x_a \\ y_a \end{bmatrix}$ added to a vector $\vec{b} = \begin{bmatrix} x_b \\ y_b \end{bmatrix}$ is the vector $\begin{bmatrix} x_a + x_b \\ y_a + y_b \end{bmatrix}$. As you can see, the addition of two vectors results in another vector. Here is a illustration to help clear up things:

You can think of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ as one unit of direction in the x and y axis respectively.

This can be extrapolated into any dimensions:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} a+d \\ b+e \\ c+f \end{bmatrix}.$$

Let us analyze Figure 2.2. Firstly, notice that $\vec{a} = 1\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\vec{b} = 1\hat{\mathbf{i}} + 1\hat{\mathbf{j}}$. Thus, $\vec{a} + \vec{b} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$, exactly like in our figure. One way to understand this is to “attach” the vector $\vec{a}$ ’s tail to $\vec{b}$ ’s head by translating $\vec{a}$. You may have seen this in your physics class before. Similarly, the subtraction of vectors $\vec{x} = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} c \\ d \end{bmatrix}$ is $\vec{x} - \vec{y} = \begin{bmatrix} a-c \\ b-d \end{bmatrix}$. Of course, we can use the same “head to tail” model to explain vector subtraction. Subtracting $\vec{y}$ from $\vec{x}$ is simply finding the vector whose tail is on the head of $\vec{y}$ and whose head is on the head of $\vec{x}$. However, some of you may be scratching your heads right now, what if a vector does not start from the *origin* (or the coordinate $(0, 0)$) but its tail lands on, say $(1, 1)$. How do we express such a vector? The answer may be surprising, both vectors have the same representation. However, unless specifically stated, *all* vectors start from the origin. This is also why the “head to tail” model works, while the “tail” of the vector is not at the origin, translating it to the origin results in an *identical* vector.



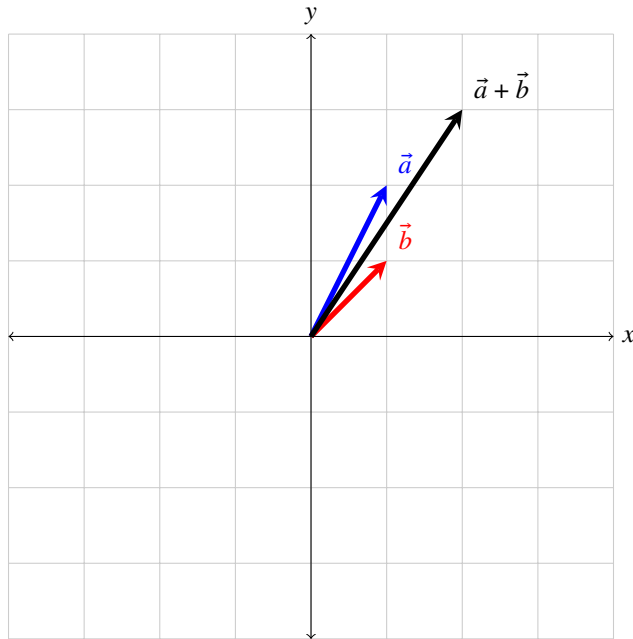


Figure 2.2 Illustration of vector addition.

Example 2.1 Addition of vectors

Here are some examples of vectors:

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix} \quad \vec{c} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

Here are some examples of adding and subtracting vectors. Note that addition are *commutative*, in other words the order does not matter:

$$\vec{a} + \vec{b} = \begin{bmatrix} 1 - 3 \\ 2 + 4 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$\vec{a} - \vec{c} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$\vec{a} + \vec{b} - \vec{c} = \begin{bmatrix} -2 \\ 6 \end{bmatrix} - \vec{c} = \begin{bmatrix} -4 \\ 8 \end{bmatrix}$$



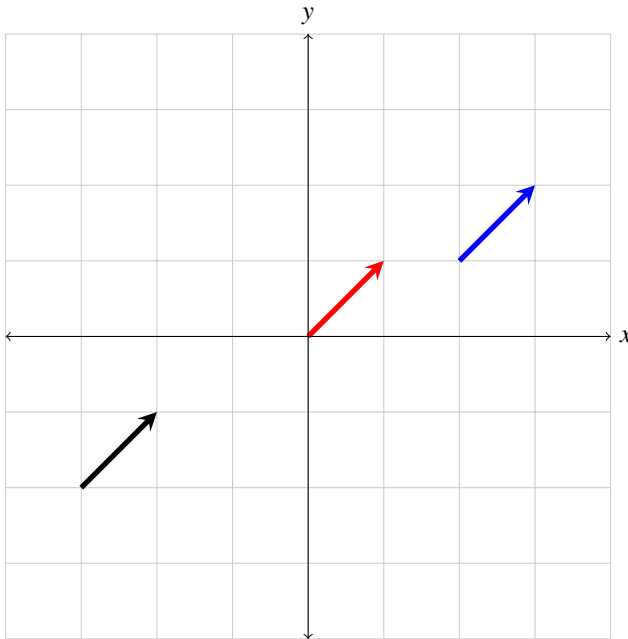


Figure 2.4 Three identical vectors.

Given a vector $\vec{a}$, $-\vec{a}$ can be thought of the same vector with the same magnitude but pointing in the *opposite* direction. To find it, simply flip the signs of all the components in $\vec{a}$. If you add $\vec{a}$ to $-\vec{a}$, you get what is called the *zero vector*, or the vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. This is a vector without direction has zero magnitude. We shall denote it $\vec{0}$. Lastly, I want to briefly touch on the idea of *magnitude* which is basically the length of a vector. The derivation of the following formula is simply the Pythagorean theorem.

Definition 2.3 Magnitude of vector

The magnitude of a vector $\vec{v}$ in n-dimensional space is as such:

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

A *unit* vector is a vector with magnitude 1.

Try to attempt the following exercise:

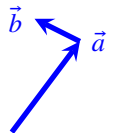


Figure 2.5: Visualisation of “head to tail” addition of $\vec{a}$ and $\vec{b}$.

2.1.1 Exercise

Problems

- Given that $\vec{a} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $\vec{c} = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$, calculate:
 - $\vec{a} + \vec{b}$
 - $\vec{a} - \vec{c}$
 - $\vec{a} - \vec{b} - \vec{c}$
- Are the following statements true or false:
 - The subtraction of two vectors *always* results in a vector.
 - When any vector is added to the zero vector, the result *always* is not a zero vector.
- Find the magnitude of $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$.
- (Difficult)** Prove that the addition of vectors is *associative* given that the addition in $\mathbb{R}$ is associative.²

² An operation $\circ$ is associative if order in which it is applied does not matter, in this case $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$.

Solutions

- $\begin{bmatrix} 1 \\ 5 \end{bmatrix}$
 - $\begin{bmatrix} 6 \\ 6 \end{bmatrix}$
 - $\begin{bmatrix} 8 \\ 7 \end{bmatrix}$
- True since subtraction can be viewed as $\vec{a} + (-\vec{b})$ and $-\vec{b}$ is a vector.
 - False since $\vec{0} + \vec{0} = \vec{0}$.
- 5 and $\sqrt{50}$.

4. Let $\vec{x}, \vec{y}, \vec{z}$ be vectors in $\mathbb{R}^n$ (N-dimensional euclidean space). Notice that:

$$\begin{aligned}(\vec{x} + \vec{y}) + \vec{z} &= \left(\begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} \right) + \begin{bmatrix} z_1 \\ \dots \\ z_n \end{bmatrix} \\&= \begin{bmatrix} x_1 + y_1 + z_1 \\ \dots \\ x_n + y_n + z_n \end{bmatrix} \\&= \begin{bmatrix} x_1 + (y_1 + z_1) \\ \dots \\ x_n + (y_n + z_n) \end{bmatrix} \\&= \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} + \left(\begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} + \begin{bmatrix} z_1 \\ \dots \\ z_n \end{bmatrix} \right) \\&= \vec{x} + (\vec{y} + \vec{z})\end{aligned}$$

2.2 SCALING AND DOT PRODUCT

In this chapter we will deal two kinds of multiplication with vectors. By far the easiest would be scaling a vector. As we have previously discussed, the magnitude of a vector is the length of a vector. But what if I want to “elongate” the vector by some amount to make it longer or shorter. This is called *scaling*.

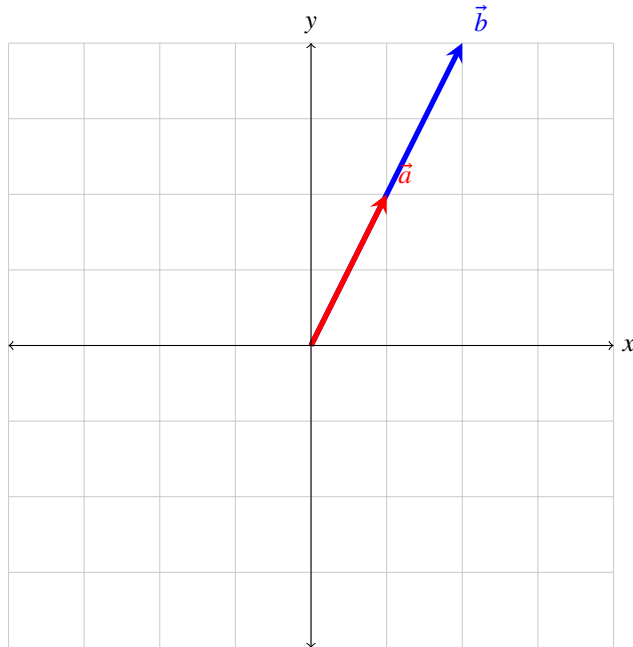


Figure 2.6 Two vectors, one twice as long as the other.

Take a look at Figure 2.6, $\vec{b}$ is twice as long as $\vec{a}$. What should we multiply to $\vec{a}$ to make as long as $\vec{b}$? If you guessed 2, then you are right. (Not sure what else you could have guessed.) To multiply a vector by a constant, we multiply each individual component of the vector by the constant. From the figure, we can identify that $\vec{a} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ so $2\vec{a} = \begin{bmatrix} 2 \times 2 \\ 2 \times 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ which is equal to $\vec{b}$. You may extrapolate this to higher dimensions.

Definition 2.4 Scaling a vector

Scaling a n -dimensional vector $\vec{v}$ by a constant n is defined as such:

$$n\vec{v} = n \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} n \times v_1 \\ \vdots \\ n \times v_n \end{bmatrix}$$

Never ever write $n \times \vec{v}$ or $\vec{n}v$ or you will be cursed into the thundering depths of careless *mistakes* everywhere.



Let us try something fun, what if we multiplied the vector by a negative number? Well, the computation remains the same. As you can see from Figure 2.7, $\vec{a} = (-0.5)\vec{b}$. You may wish to compute this yourself to verify. One thing to note here is that the vector gets flipped across the origin when the constant the vector is multiplied to is negative.

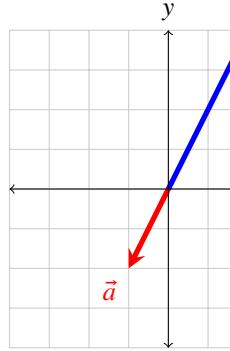


Figure 2.7: Two vectors, one is scaled by -0.5.

Challenge 2.1 Scaled vector magnitude

Also, when scaling a vector by some constant, the reader may want to verify using ?? 2 that the magnitude of the vector is also scaled by that constant. This should be a trivial proof.



You may have noticed that I have abstained from using either the $\cdot$ or $\times$ operators. The reason is bad notation which is sprinkled throughout mathematics. These two operators have special meanings in the realm of linear algebra. We shall first explore the easier one, the *dot product*.

Dot products use the $\cdot$ operator and only acts on vectors, do not abuse notation by using the $\cdot$ operator in the context of scaling a vector. Dot products are actually fairly straightforward to compute:

Definition 2.5 Dot product

The dot product of two vectors in n -dimensional space $\vec{a}$ and $\vec{b}$ is:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = (||\vec{a}||)(||\vec{b}||) \cos \theta$$

where θ is the angle between the two vectors. We shall refer to the formula between the first and second equal signs the “first definition of dot products” and the formula after the second equal sign the “second definition of dot products”.



From Section 2.2, we can tell that the dot product tells us something about the relationship between the two vectors. In essence, if the dot product between two vectors is large and positive, then the vectors are pointing in roughly the same direction. Else, if the product is very negative, then the dot product is very negative.

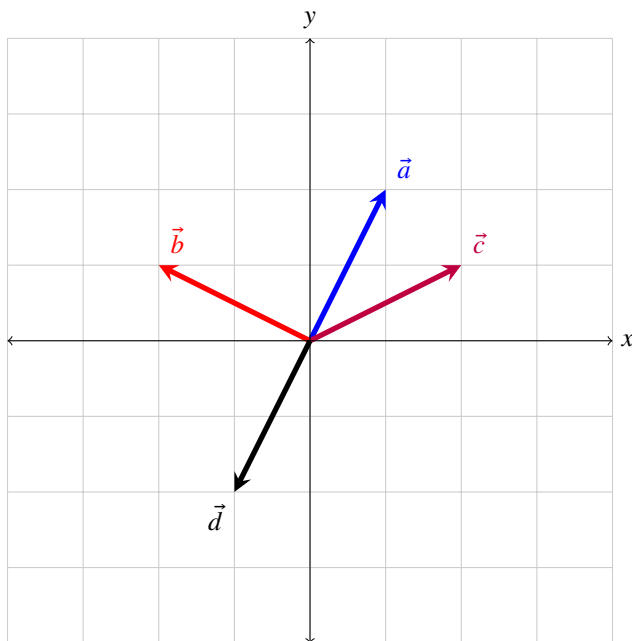


Figure 2.8 Four vectors pointing in different directions.

Firstly, note down the components of each vector in Figure 2.8. Then, let us compute $\vec{a} \cdot \vec{b}$. By Section 2.2, it is $1(-2) + 2(1) = 0$. Also, it is important to take note that these two vectors are orthogonal (at right angles with each other). This brings us to an important property of dot products, if two vectors are orthogonal, then their dot product is 0. This can be seen from the fact that $\cos\left(\frac{\pi}{2}\right) = 0$ and from the second definition of dot products in Section 2.2. In a sense, the dot product tells you how aligned are the directions of two vectors. Another example would be taking the dot product of two identical vectors, such as $\vec{a} \cdot \vec{a}$. The dot product there is $1(1) + 2(2) = 5$. In addition, we know $\cos 0 = 1$ so according to the second definition, $\vec{a} \cdot \vec{a} = (||\vec{a}||)(||\vec{a}||) = (||\vec{a}||)^2$. If we take the square root on both sides, $||\vec{a}|| = \sqrt{\vec{a} \cdot \vec{a}}$ which can be rewritten as $||\vec{a}|| = \sqrt{a_1^2 + \dots + a_n^2}$. Drum roll please. This is exactly the formula we got in ?? 2 for the magnitude of a vector. This is a pretty nice result.

Example 2.2 Calculating dot products

From Figure 2.8, we can see that:

$$\vec{a} \cdot \vec{c} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 1(2) + 2(1) = 4$$

$$\vec{a} \cdot \vec{d} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -2 \end{bmatrix} = 1(-1) + 2(-2) = -5$$

**Remark 2.1**

Generally, when finding the dot product, we use the first form in Section 2.2 as it is typically easier to compute. It is also important to know that the dot product of two vectors results in a number, not a vector like in scaling.

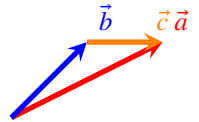


Figure 2.9: Figure of a triangle formed by three vectors.

1. If $\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, compute the following:

- (a) $2\vec{x}$
- (b) $\vec{x} \cdot \vec{y}$
- (c) $\vec{y} \cdot \vec{x}$

2. Prove or disprove that the dot product *commutes*.³

1. Given two vectors $\vec{a}$ and $\vec{b}$ separated by an angle θ , they form a triangle with a third side $\vec{c} = \vec{a} - \vec{b}$ (See Figure 2.9). By taking the dot product of $\vec{c}$ and itself, derive the law of cosines. (Hint: $(\vec{x} \cdot \vec{y}) \cdot (\vec{z} \cdot \vec{w}) = \vec{x} \cdot \vec{z} + \vec{x} \cdot \vec{w} + \vec{y} \cdot \vec{z} + \vec{y} \cdot \vec{w}$)

2. (a) $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$
 (b) 11
 (c) 11

3. The dot product commutes. From the second definition of the dot product from Section 2.2, we can see that $\vec{a} \cdot \vec{b} = (||\vec{a}||)(||\vec{b}||) \cos(\theta) = (||\vec{b}||)(||\vec{a}||) \cos(\theta) = \vec{b} \cdot \vec{a}$. Therefore, the dot product commutes.

³ An operation $\circ$ is commutative if $x \circ y = y \circ x$.

4. Let a, b, c be the magnitudes of the respective vectors. Just take the dot product of $\vec{c}$ and itself. ⁴

$$\begin{aligned}\vec{c} \cdot \vec{c} &= (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) \\ &= a^2 - 2\vec{a} \cdot \vec{b} + b^2 \\ &= a^2 + b^2 - 2ab \cos(\theta)\end{aligned}$$

Which is exactly the law of cosines.

⁴ Note that $\vec{a} \cdot \vec{a} = (||\vec{a}||)^2$ as evident from the second definition of Section 2.2.

2.3 MATRICES

In this chapter, we will introduce a new mathematical object that you have probably heard before but not sure what it is. To be honest, many mathematics students only know how to *compute* matrix operations but does not know the meaning behind them. I will try to do my best to explain them in a clear manner and I hope that you would appreciate matrices more for their beautiful geometric meaning than their mind-numbing calculations. A great place to start is the *basis vectors* introduced back in Section 2.1. All vectors can be written as a unique linear combination of the basis vectors. But $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are not the only basis vectors out there, what if I wanted to describe vectors using another set of basis vectors? Perhaps that would be more convenient for my special use case. Well, it is actually pretty easy to do so. For the sake of this argument, let us try to rotate all of the vectors in a normal 2d vector field anti-clockwise by 90° . Notice how we could not describe $\vec{b}$ as some constant multiplied to $\vec{a}$ in Figure 2.10. We need some other tool to help us. If we change the basis vectors of $\vec{a}$ to another pair of basis vectors, can we somehow get $\vec{b}$? Here is a thought: replace the $\hat{\mathbf{i}}$ with $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\hat{\mathbf{j}}$ with $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$. So, $\vec{a} = 1\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ will become $\vec{a} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 2\begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$. This is pretty intriguing, by changing the basis vectors, we have basically “morphed” $\vec{a}$ into $\vec{b}$. This “morphing” is called a *linear transformation* or *linear map*. In some of the visualisations of the linear transformations I might keep the original grid in the background in a lighter color.

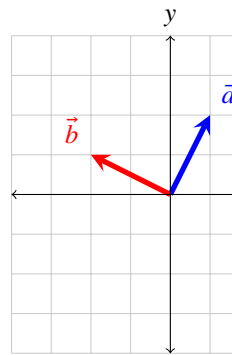


Figure 2.10: Two vectors, one rotated anti-clockwise by 90° .

Example 2.3 Linear transformations

Here are some examples of linear transformations, the blue grid is the vector space after the transformation. Take note where the basis vectors are after the transformations.

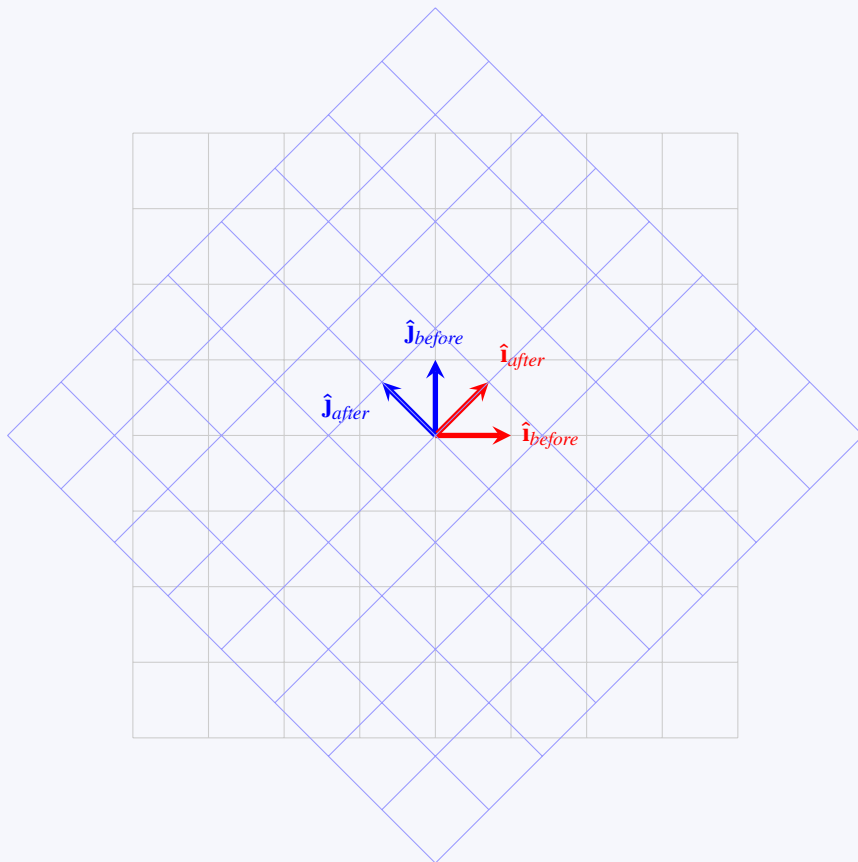
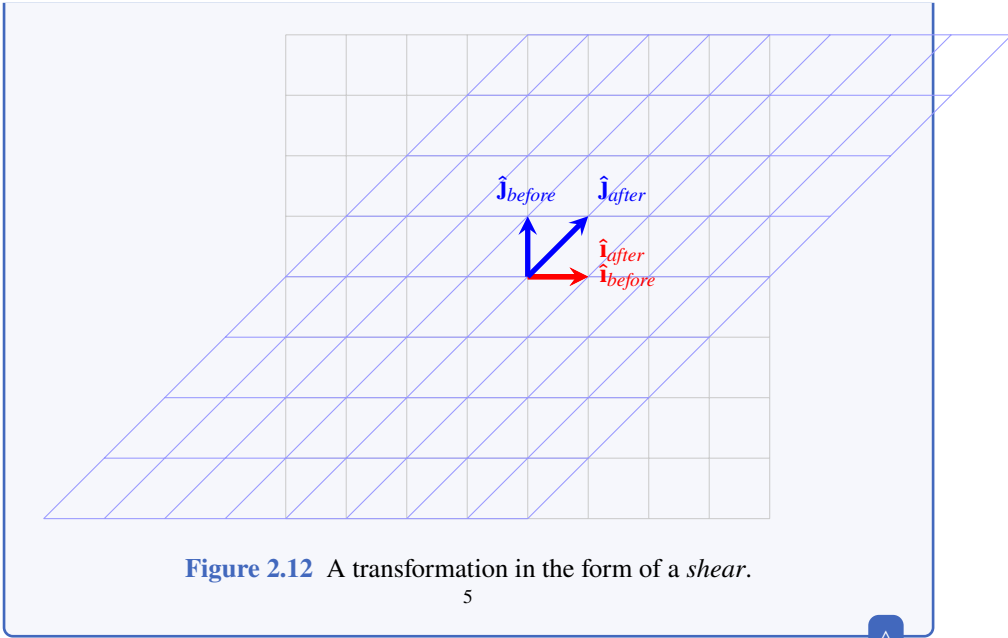


Figure 2.11 A transformation in the form of an anti-clockwise *rotation*.



5

To describe these transformations, we actually only need to describe where the basis vectors land after the transformation. No other information required. We present the location of the new basis vectors in the form of a *matrix*:

⁵ In Figure 2.12, the $\hat{i}$ vector remains at the same location.

Definition 2.6 Matrices

In linear algebra, matrices generally represent linear transformations. A $n \times m$ matrix is one with n rows and m columns. The n th column of the matrix denotes the representation of the corresponding basis vector after the linear transformation.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

represents a matrix that moves the $\hat{i}$ basis vector to $\begin{bmatrix} a \\ c \end{bmatrix}$ and moves the $\hat{j}$ basis vector to $\begin{bmatrix} b \\ d \end{bmatrix}$.

Remark 2.2

Matrices can also represent a variety of things apart from linear transformations. It is just a mathematical object representing a bunch of things arranged in rows and columns.



It should go without saying that any vector in the original vector space will get moved after a transformation. To calculate where a vector ends up after a transformation, we multiply the matrix by the vector. This is where many textbooks make linear algebra feel laborious. In order to make the process more intuitive, recall that a vector has a $\hat{i}$ and $\hat{j}$ component. Now, we swap the basis vectors for the new basis vectors described in our matrix. This is just multiplying a vector by a constant, which you have done before.

Example 2.4 Deriving a linearly transformed vector

Take the transformation in Figure 2.12 which is represented by the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Let us say in the original vector space, there is a vector $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Then, the vector's representation after the transformation:

$$\begin{aligned} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} &= 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{aligned}$$

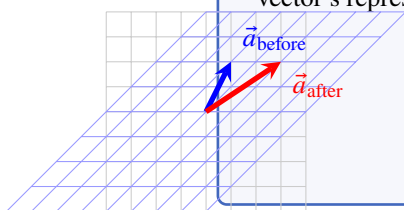


Figure 2.13: Figure of a vector before and after a shear transformation like in Figure 2.12.

If you have come from a linear algebra course, this may feel very refreshing. Instead of memorising what needs to be multiplied, you just swap the basis vectors for some other basis vectors. While we get the representation of $\vec{a}$ from Figure 2.13, it also important to take note that in the new vector space after the transformation, $\vec{a}$ has one unit of direction in the horizontal direction and two units of direction in the “vertical” direction. This is just like the original representation of $\vec{a}$, just that the vector space has been sheared. This change of basis vector is important enough to have its own name, *change of basis*.

With this knowledge, let us proceed to one of the most hated parts of linear algebra, *matrix multiplication*. These few words can bring out so much frustration out of so many students. It is quite a laborious and boring process if you do not grasp the meaning and motivation behind it. In actuality, the idea is simple. When you multiply two matrices together, you are effectively applying two linear transformations, one after another. The result is also a matrix, which is a linear transformation, hence the operation is *closed*.

This is only true if the matrix is $n \times n$, which are the ones we are dealing with.

Example 2.5 Deriving matrix multiplication

Say you have two matrices, a 90° anti-clockwise rotation $x = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and a horizontal shear $y = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. xy is simply applying the shear first before the rotation. (The right to left reading is similar to that of function composition)

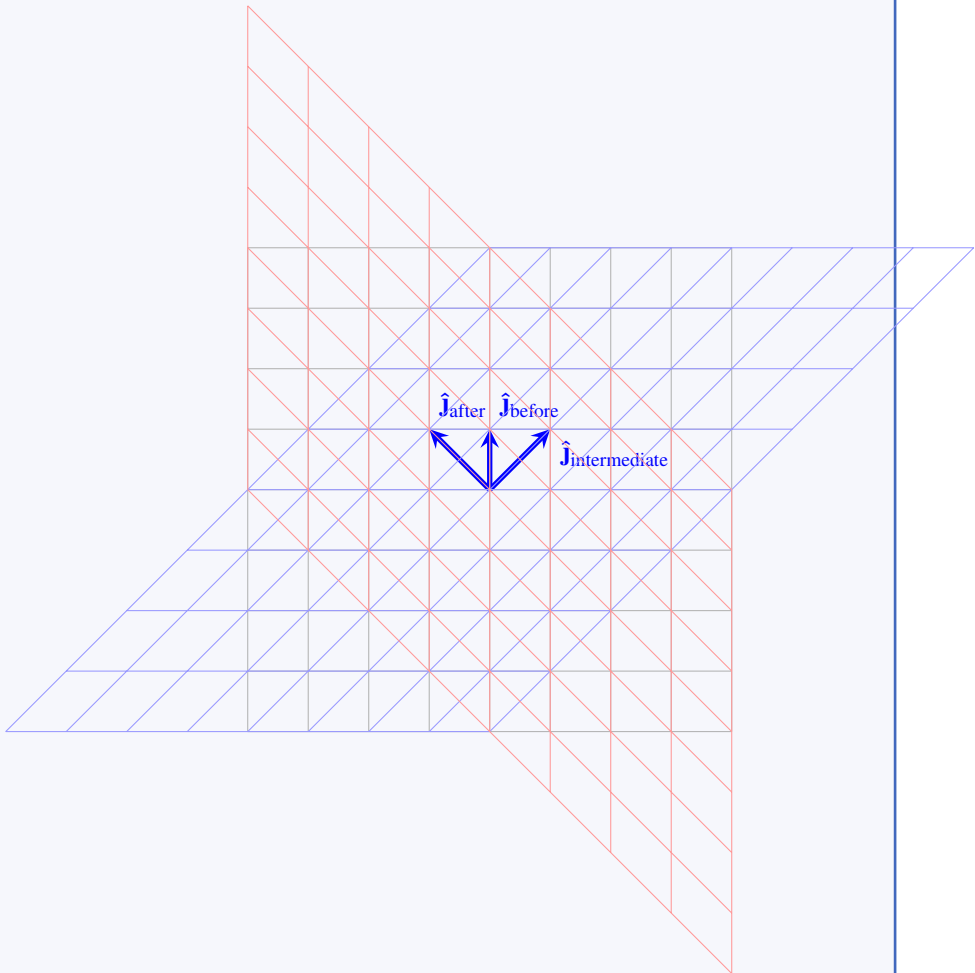


Figure 2.14 Figure of applying two linear transformations.



The identity matrix is simply the transformation that leaves the vector space unchanged.

It should not be difficult to see that the end result of applying the two linear transformations in the previous examples resulted in another linear transformation (the red grid). We wish to find the representation of this linear transformation as a matrix. Let us go through this step by step. First of all, the basis vectors of the *identity matrix*, gets moved to some new location. In other words, given a matrix $a = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$, we know that the basis vectors gets moved to $\begin{bmatrix} a_1 \\ a_3 \end{bmatrix}$ and $\begin{bmatrix} a_2 \\ a_4 \end{bmatrix}$ respectively for the $\hat{i}$ and $\hat{j}$ vectors. Now, we have another transformation $b = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$. To know where the $\hat{i}$ vector lands after both transformations, notice that it is simply multiplying the intermediate position of it by b , which is $\begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_1 b_1 + a_3 b_2 \\ a_1 b_3 + a_3 b_4 \end{bmatrix}$. A similar line of reasoning may be used for $\hat{j}$. Hence, the result is:

$$ba = \begin{bmatrix} a_1 b_1 + a_3 b_2 & a_2 b_1 + a_4 b_2 \\ a_1 b_3 + a_3 b_4 & a_2 b_3 + a_4 b_4 \end{bmatrix}$$

This is the reason many students find linear algebra tedious. Most courses make students memorise the formula which is extremely rote. I wish that you understand what is going on behind the scenes on this formula. With that said, it is best to get some experience with these multiplications:

Example 2.6 Matrix multiplication

Given that $x = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $y = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $z = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$,

$$xy = \begin{bmatrix} 5 & 11 \\ 11 & 25 \end{bmatrix}$$

$$yx = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$$

$$x(yz) = \begin{bmatrix} 102 & 118 \\ 230 & 266 \end{bmatrix}$$

$$(xy)z = \begin{bmatrix} 102 & 118 \\ 230 & 266 \end{bmatrix}$$

This examples shows that matrix multiplication does not commute but it is associative.



1. If $\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $a = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ and $b = \begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix}$, compute:

(a) $a\vec{x}$

(b) ab

(c) $ab\vec{x}$ 2. **(Difficult)** Show that matrix multiplication is *associative* for all 2×2 matrices.

1. (a) $\begin{bmatrix} 8 \\ 14 \end{bmatrix}$

(b) $\begin{bmatrix} 36 & 41 \\ 64 & 37 \end{bmatrix}$

(c) $\begin{bmatrix} 118 \\ 210 \end{bmatrix}$

2. Let $x = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ and define y and z likewise, notice that: ⁶

$$\begin{aligned}
 x(yz) &= \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} \begin{bmatrix} z_1y_1 + z_3y_2 & z_2y_1 + z_4y_2 \\ z_1y_3 + z_3y_4 & z_2y_3 + z_4y_4 \end{bmatrix} \\
 &= \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \quad \text{let } a=yz \\
 &= \begin{bmatrix} a_1x_1 + a_3x_2 & a_2x_1 + a_4x_2 \\ a_1x_3 + a_3x_4 & a_2x_3 + a_4x_4 \end{bmatrix} \\
 &= (xy)z
 \end{aligned}$$

⁶ The derivation for $(xy)z$ will be left as an exercise for the reader. Both should evaluate to the same expression.

2.4 DETERMINANTS, INVERSES AND CRAMER'S RULE

At this juncture, you should be familiar with the concept that a matrix represents a linear transformation and it morphs the vector space. Also, if you imagine the basis vectors of any vector space as lengths of a parallelogram, then you might want to find the area of that parallelogram. For example, the identity matrix's basis vectors forms a square with area 1. Likewise, given a matrix $x = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we can display it as:

To find the area of A in Figure 2.15, we may use the sine rule (let $\vec{u} = \begin{bmatrix} a \\ c \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} b \\ d \end{bmatrix}$):

$$\begin{aligned}
 A &= (||\vec{u}||)(||\vec{v}||) \sin(\theta) \quad \text{where } \theta \text{ is the angle between the vectors} \\
 &= (||\vec{u}'||)(||\vec{v}||) \cos(\theta') \quad \text{where } \vec{u}' \text{ is } \begin{bmatrix} a \\ -c \end{bmatrix}, \text{ perpendicular to } \vec{v} \text{ and } \theta' \text{ is } 90^\circ \\
 &= \begin{bmatrix} -c \\ a \end{bmatrix} \cdot \begin{bmatrix} b \\ d \end{bmatrix} \quad \text{by Section 2.2, definition of dot product} \\
 &= ad - bc
 \end{aligned} \tag{2.4.1}$$

The resultant area is also known as the *determinant* of a matrix.

Definition 2.7 Determinant of 2×2 matrix

The determinant of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is represented as:

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

and it represents the signed area of a parallelogram with sides of the basis vectors described in A . Meanwhile, a 3×3 matrix $B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ has a determinant of:

$$\begin{aligned}
 \det(B) &= \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} \\
 &= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}
 \end{aligned}$$

and it represents the volume of the parallelepiped formed by the three basis vectors in B .



Example 2.7 Finding determinants

Given that $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$:

$$\begin{aligned}\det(A) &= 1 \times 4 - 2 \times 3 \\ &= -2\end{aligned}$$

$$\det(B) = 0 \quad (2.4.2)$$

7

$$\begin{aligned}\det(C) &= 1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= (5 \times 9 - 8 \times 6) - 2(4 \times 9 - 6 \times 7) + 3(4 \times 8 - 7 \times 5) \\ &= 0\end{aligned}$$



Next, we would move onto the *inverse* of a matrix. Think of a number x , $x^{-1} \times x = 1$, 1 is also called the multiplicative identity. Likewise, given a matrix Y , $YY^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ which is the identity matrix. If you think of Y as a transformation, then Y^{-1} “undo” the transformation of Y .

7 Equation (2.4.2) shows a more general property, given that $X = \begin{bmatrix} a & b \\ a & b \end{bmatrix}$, $\det(X) = 0$. The proof is trivial and shall be left as a verification to the reader.

Definition 2.8 Inverse of a matrix

The (multiplicative) inverse of a matrix X is denoted X^{-1} where:

$$XX^{-1} = I$$

where I is the identity matrix. If X has an inverse, it is called an *invertible* matrix. X has an inverse if and only if $\det(X) \neq 0$. To find the inverse of a matrix $Y = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we use the following formula:

$$Y^{-1} = \frac{1}{\det(Y)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (2.4.3)$$

8



8 When multiplying a matrix by a constant, you multiply each entry in matrix by the constant.

Example 2.8 G

ven $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix}$:

$$\begin{aligned} A^{-1} &= \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix} \end{aligned}$$

$\det(B) = 0$, hence B is not invertible.



Until now, this chapter has been focusing mainly on computation. The reader may feel that these computations are excessively dull and boring. That being said, here is a application of determinants, *Cramer's rule*. Cramer's rule may be used to solve a system of linear equations. Firstly, we will represent a simultaneous equation as a product of vectors and matrices. Given a simultaneous equation:

$$\begin{aligned} x + y &= 10 \\ 2x - y &= 14 \end{aligned} \tag{2.4.4}$$

We write it as:

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \end{bmatrix} \tag{2.4.5}$$

This is also called the “matrix form” of a simultaneous equation. It also important to recognize that Equation (2.4.5) is in the form of $A\vec{v} = B$, it is easy to see that $\vec{v} = BA^{-1}$. From Equation (2.4.3), we can find BA^{-1}

$$\begin{aligned} BA^{-1} &= \begin{bmatrix} 10 \\ 14 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 2 \end{bmatrix} \end{aligned}$$

This is also the solution for Equation (2.4.5), x is indeed 8 and y is indeed 2. However, this method might not be the best. If there were three unknowns and three equations, how could you solve the equations? We may apply Cramer's rule. Consider a system of n linear equations with n unknowns $(x_1, x_2, \dots, x_n)$, it can be represented as $Ax = B$ as we have seen before. The solutions for the equations are:

$$x_i = \frac{\det(A_i)}{\det(A)} \quad i = 1, \dots, n$$

Where x_i is the i -th solution and A_i is the matrix formed by replacing the i -th column of A by B .

Example 2.9 Solving linear equations with Cramer's rule

Given the following system of linear equations of three unknowns:

$$x_1 + x_2 + x_3 = 12$$

$$x_1 - x_2 + x_3 = 4$$

$$x_1 - x_2 - x_3 = -6$$

and its matrix form is:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 4 \\ -6 \end{bmatrix}$$

The solutions are:

$$\begin{aligned} x_1 &= \frac{\det(A_1)}{\det(A)} \\ &= \frac{\det \left(\begin{bmatrix} 12 & 1 & 1 \\ 4 & -1 & 1 \\ -6 & -1 & -1 \end{bmatrix} \right)}{\det \left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix} \right)} \\ &= \frac{12}{4} \\ &= 3 \end{aligned}$$

$$x_2 = \frac{\det(A_2)}{\det(A)}$$

$$\begin{aligned}
 & \det \begin{pmatrix} 1 & 12 & 1 \\ 1 & 4 & 1 \\ 1 & -6 & -1 \end{pmatrix} \\
 &= \frac{\det \begin{pmatrix} 1 & 12 & 1 \\ 1 & 4 & 1 \\ 1 & -6 & -1 \end{pmatrix}}{\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}} \\
 &= \frac{16}{4} \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 x_3 &= \frac{\det(A_3)}{\det(A)} \\
 &= \frac{\det \begin{pmatrix} 1 & 1 & 12 \\ 1 & -1 & 4 \\ 1 & -1 & -6 \end{pmatrix}}{\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}} \\
 &= \frac{20}{4} \\
 &= 5
 \end{aligned}$$



As you can tell, Cramer's rule is very powerful. In the real world, however, *gaussian elimination* is a much faster alternative than Cramer's rule. It is also unlikely for the reader to be computing large matrices determinants by hand, we have computers for that. This also applies to matrix multiplication. This book, and more generally calculus courses only need knowledge on computing determinants of up to 3×3 matrices.

2.4.1 Exercise

Problems

1. Find the determinants and inverses, if any, for the following matrices:

(a) $\begin{bmatrix} 1 & 9 \\ 2 & 10 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 4 \\ 7 & 9 \end{bmatrix}$

2. Find the determinant of the following matrix:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 3 & 6 \end{bmatrix}$$

3. Solve the following simultaneous equations using matrices:

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 \\ x_1 + 2x_2 + x_3 &= 4 \\ x_1 + 2x_2 + 2x_3 &= 6 \end{aligned}$$

Solutions

1. (a) -8 and $\begin{bmatrix} -\frac{10}{8} & \frac{9}{8} \\ \frac{1}{4} & -\frac{1}{8} \end{bmatrix}$

(b) 0 and no inverse

(c) -1 and $\begin{bmatrix} -9 & 4 \\ 7 & -3 \end{bmatrix}$

2. 1

3. Use Cramer's rule and $x_1 = 0$, $x_2 = 1$, $x_3 = 2$. Example for x_1 :

$$\begin{aligned} x_1 &= \frac{\det(A_1)}{\det(A)} \\ &= \frac{\begin{vmatrix} 3 & 1 & 1 \\ 4 & 2 & 1 \\ 6 & 2 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{vmatrix}} \\ &= \frac{0}{1} \end{aligned}$$

$$= 0$$

Example for x_2 :

$$\begin{aligned} x_2 &= \frac{\det(A_2)}{\det(A)} \\ &= \frac{\begin{vmatrix} 1 & 3 & 1 \\ 1 & 4 & 1 \\ 1 & 6 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{vmatrix}} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

2.5 CROSS PRODUCTS AND TRIPLE PRODUCTS

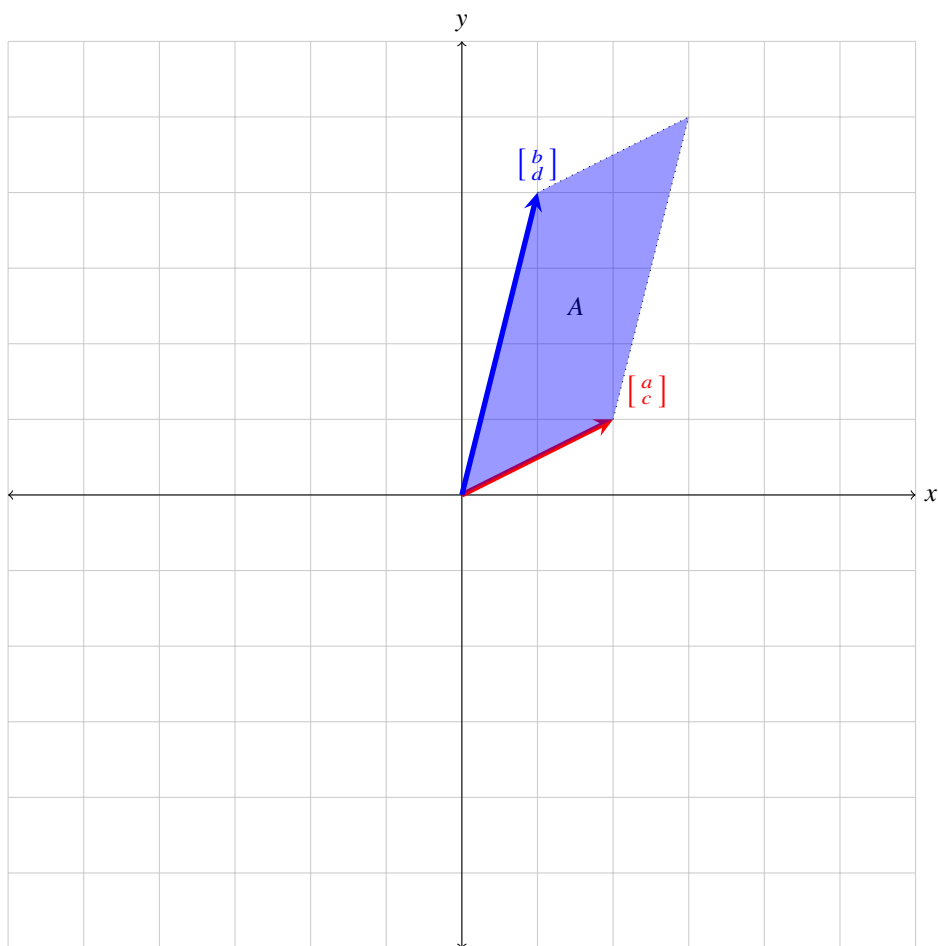


Figure 2.15 Figure of area A formed by a pair of basis vectors.